



TRACKER

Surface Inspection Light

OPERATING MANUAL



TRACKER

Issue #505

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Doc. #98-1240





WARNING: READ BEFORE OPERATING UNIT

Operation

- The **TRACKER** operates at **LETHAL High Voltages** inside the lamp and electronic compartments. Disconnect unit from power supply before replacing lamp.
- Viewing of this light by people with photosensitive epilepsy may trigger epileptic seizures.
- Eye damage may result from direct viewing of the ultraviolet lamp used in this unit.

Product recycling and disposal

This unit must be recycled or discarded according to applicable local and national regulations. Unilux encourages owners of industrial equipment to responsibly recycle their equipment when it is no longer needed. Unilux offers a variety of product return programs and services in several countries to assist equipment owners in their recycling efforts



Note:

This mark applies only to countries within the European Union (EU) and Norway. Appliances are labeled in accordance with European Directive 2002/96/EC concerning waste electrical and electronic equipment (WEEE). The Directive determines the framework for the return and recycling of used appliances as applicable throughout the European Union. This label is applied to various products to indicate that the product is not to be thrown away, but rather reclaimed upon end of life per this Directive. In accordance with the European WEEE Directive, electrical and electronic equipment (EEE) is to be collected separately and to be reused, recycled, or recovered at end of life. Users of EEE with the WEEE marking per Annex IV of the WEEE Directive, as shown above, must not dispose of end of life EEE as unsorted municipal waste, but use the collection framework available to customers for the return, recycling, and recovery of WEEE. Customer participation is important to minimize any potential effects of EEE on the environment and human health due to the potential presence of hazardous substances in EEE. For proper collection and treatment, contact your local Unilux representative.

-----NOTICE-----

The information contained in this manual was current at the time of publishing. The manufacturer reserves the right to revise and improve its product, therefore, all specifications are subject to change without notice.

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WELCOME TO THE TRACKER.....

The **TRACKER** is a part of Unilux's Spotlight series of compact high intensity surface inspection lights.

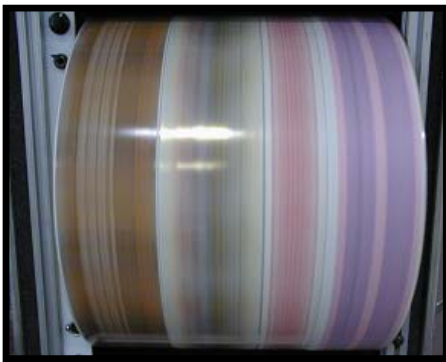
The **TRACKER** has the following features:

- Self contained design - no separate high voltage power supply.
- Operates from a 12 volt DC power source.
- Long lamp life.
- Accepts remote synchronization [trigger] signals.
- A large, bright, easy to read LED display.
- Portable or fixed mounting.
- Totally enclosed case.
- Superior design reflector offers uniform area illumination.
- X2 ÷ 2 Harmonic buttons
- Phase Control of External Trigger signal
- Low battery indicator

GETTING STARTED

STOPPING THE BLUR

The Unilux ***Tracker*** uses stroboscopic lighting to “freeze action”. When objects move at high speed, the human eye cannot focus on a single image. As a result, the eye can only see "blur". But strobe lighting stops the blur. By flashing with high intensity for extremely short durations, a strobe light will "freeze" objects traveling at speeds that cause blur. [See Illustrations Below]



WITHOUT STROBE
[BLUR ACTION]



WITH STROBE
[FREEZE ACTION]

During these short flashes, an object moving at 600 meters per minute moves only 0.2mm - so it appears to be stationary since the eye captures and retains only brief visual images. This result makes detailed inspection possible. No other type of light provides this kind of clarity and definition of surfaces in motion.

SHORT FLASH, HIGH DEFINITION

The length (duration) of time that the light is actually on for (flash) and its brightness in respect to ambient light, affect clarity. The shorter the flash duration, the sharper the image. The more the light from the strobe dominates the ambient light the better. Unilux recommends strobe lighting levels four times brighter (4x) than the ambient lighting.

WHITE LIGHT

Unilux inspection lights emit an excellent white light approaching sunlight that reveals true colors. The light's scientific color temperature is 6500 Kelvin.

HOW TO SET THE FLASH RATE

Flash rate should be adjusted within these limits for non-synchronous inspection.
[*Note: In pattern inspection, flash rate must be set to synchronize*].

- Adjust flash rate to exceed 50 flashes per second (f/s) (3000 per minute). Rates below 50 f/s can be dangerous and disruptive to personnel affected in the immediate area [See **WARNING** on first page]. Therefore, 50 f/s should be your minimum flash rate.
- Flash rates should not allow more than 15cm to 30cm of material to pass between flashes. Therefore, to calculate the maximum rate, divide the line speed in centimeters per second by 15, and this will be the maximum flash frequency in flashes per second [f/s].

Experience gained from many industries using non-synchronous inspection lighting shows that flash rates at or near 70 f/s [4200 flashes per minute, f/m] work best.

I. INTRODUCTION & APPLICATION

The Unilux **TRACKER** is a compact industrial surface inspection spotlight, specifically developed for narrow web and machine vision inspection applications.

The **TRACKER** provides narrow beam spot illumination of a flat surface covering a 25cm x 25cm area. This illuminated area results when using the **TRACKER** at its intended one foot distance.

Larger areas can be illuminated by using the **TRACKER** at greater distances, however, this will result in a corresponding reduction of light at the surface being inspected. [see **PHOTOMETRY**, page 14]

Other Unilux products are better for illuminating areas larger than the capacity of the **TRACKER**. Contact your Unilux representative for complete product information.

External and Internal Trigger Mode

The flash rate of the **TRACKER** can be controlled by an external trigger signal, or the front panel encoder.

In **External Mode**, the **TRACKER** will flash synchronously with each external trigger signal event. The **TRACKER** external trigger input is an isolated input which can be driven by either a pulse [source] or open collector [sink] type signal.

In **Internal Mode**, the front panel encoder is used to set the flash rate of the **TRACKER**. The rate control sets the flash rate from a minimum of 30 f/m up to the maximum of 6000 f/m.

The **TRACKER** has a maximum flash rate of 6000 f/m. The maximum flash energy is 0.25 joules per flash for flash rates up to 3000 f/m. At higher flash rates the flash energy is automatically reduced to limit the flashlamp power to 13 watts maximum.

NOTE:

When the external trigger connector is installed, the internal rate control is disabled.

Built-in Digital Readout

The **TRACKER** contains a digital readout for displaying the flash rate in flashes per minute [f/m]. The readout is a five digit LED display for indicating the **TRACKER** flash rate when operating in **Internal Mode**. [ie. using the front panel encoder]

The display is automatically blanked whenever the **TRACKER** is used in **External Mode**. [ie. operating with external trigger]

Portable or Fixed Mounting

The **TRACKER** can be used as a hand held portable diagnostic tool when using its optional handles.

The standard handle mounts to the **TRACKER** chassis by using a single thumbscrew. The DC Power Handle uses [2] screws for its mounting. (See figures below)

Note: A full installation sheet will be supplied by Unilux with each kit.

Optional Standard Handle

UPN 05-1559



Optional Battery Handle

UPN 04-1597-110



The Battery Pack Handle is part of a kit, which includes a rechargeable DC Power Pack, a 1 hour charger, and a DC Power Handle containing a momentary pushbutton switch. The DC Power Pack provides approximately 70 minutes of continuous operation.

A special 1 hour charger is supplied with the kit, which will fully charge the DC Power Pack in a 1 hour period. [See Chapter VII. **Charging the DC Power Pack**]

The **TRACKER** can also be mounted for more permanent inspection applications by mounting it to an Articulating Swing Arm that utilizes the 1/4-20 tap on the **TRACKER** chassis. [See chapter II. **INSTALLATION**]

An AC Handle kit is available for powering the **TRACKER** for portable inspection applications. (*Note: You must select the adaptor plug listed below**)

Optional AC Handle Kit

UPN 04-1632



ADAPTER CORD

USA - 60-1178

EUROPE - 60-1118

UK - 60-1120

AUSTRALIA - 60-1124

Also an AC to DC wall plug-in power supply (12VDC @ 2A) is available for powering the **TRACKER** for continuous inspection applications.

II. INSTALLATION

The Unilux *TRACKER* can be installed in a variety of positions, using the optional mounting accessories.



Articulating Arm
45" reach
(PN 04-1633)

ARTICULATING ARM MOUNTING BRACKETS



Clamp
(PN 56-1006)

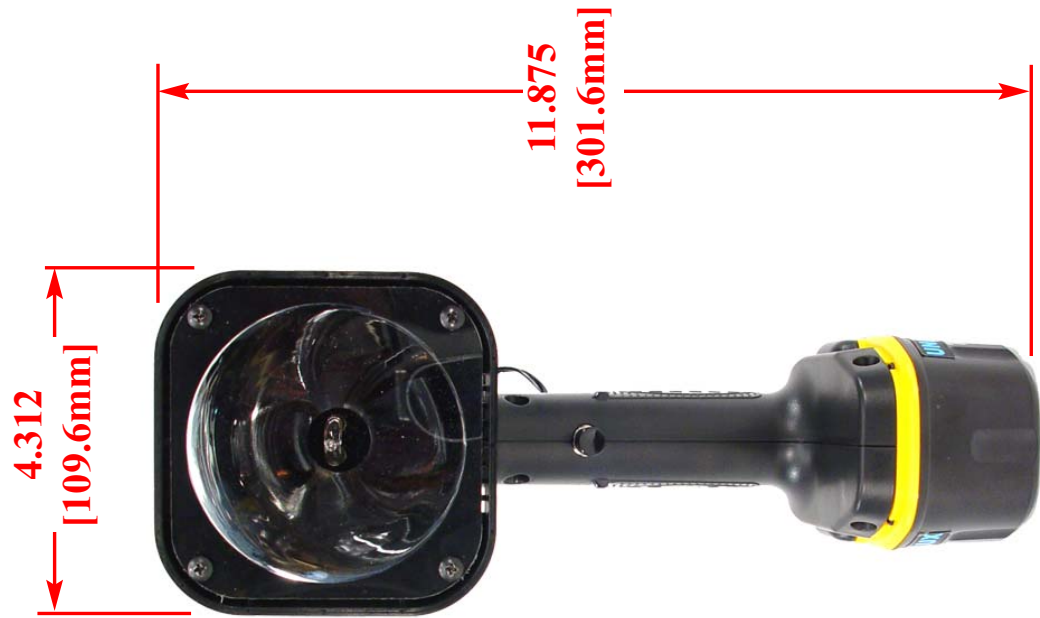


Wall
(PN 56-1007)

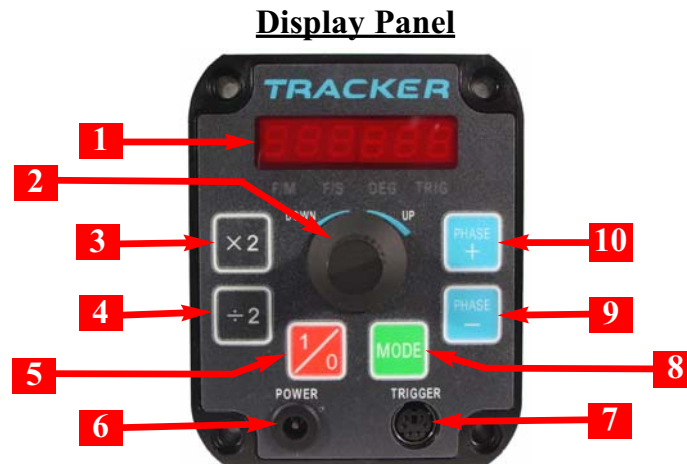


Surface
(PN 56-1008)

TRACKER Dimensional Outline



III. OPERATION



- 1 LED Display:** Indicates flash rate in flashes per minute when using Internal mode. The display is automatically blanked when using External mode.
- 2 Rate Control knob:** In internal mode, turning the Rate Control Knob clockwise increases the flash rate. Turning the Rate Control Knob counter clockwise decreases the flash rate. The size of the incremental change in the flash rate is dependent on the speed that the Rate Control Knob is changed. For fine adjustments of the flash rate turn the knob slowly. For larger changes of the flash rate, turn the knob quicker.
- 3 X2 Switch:** In internal mode, pressing the X2 button will double the flash rate. this button allows the user to make rapid large scale changes to the flash rate and aids in determining the true RPM of a rotating target
- 4 ÷2 Switch:** In internal mode, pressing the ÷2 button will halve the flash rate. This button allows the user to make rapid large scale changes to the flash rate and aids in determining the true RPM of a rotating target.
- 5 Power On/Off switch.**
- 6 Power Input connector:** 2.1mm / 5.5mm power jack for connection to low voltage power source [12VDC @ 1.5A]
- 7 Trigger Input connector:** 5 pin Mini DIN connector for input of external trigger signals.
- 8 Mode Switch:** Press to change the function of the Rate Control Knob.
- 9 Phase - Switch:** In External Mode, pressing the Phase- button decreases the phase delay of the flash relative to the incoming trigger signal. For a repeating pattern, this has the effect of moving the relative position of the target object in the field of view.
- 10 Phase + Switch:** In External Mode, pressing the Phase+ button increases the phase delay of the flash relative to the incoming trigger signal. For a repeating pattern, this has the effect of moving the relative position of the target object in the field of view.

III. OPERATION

TRACKER provides two [2] modes of operation:

1. **Internal Mode:** In this mode the rate of strobe flashing is controlled by the “RATE” control. The “RATE” control is an encoder that adjusts the flash rate from 30 f/m [flashes per minute] to the maximum flash rate of 6000 f/m.
2. **External Mode:** The external mode of operation is automatically selected whenever the prewired trigger signal connector is plugged into the *TRACKER* display panel trigger connector.
3. The phase delay will create the appearance of scrolling the viewed surface. When the phase delay is increased, the stroboscopic image appears to move forward through the illuminated area. When the phase delay is decreased, the stroboscopic image appears to move backwards through the illuminated area.

The phase delay is controlled by the “PHASE” keys. The “PHASE” keys increase or decrease the phase delay of the flash from 0° to 360° in 1.4° increments. Pressing the “Ø” key will increase the phase delay. Pressing the “Ø” key will decrease the phase delay. When the keys are released, the phase delay will remain at the last setting.

An external trigger signal can flash the lamp synchronously at any flash rate up to the maximum flash rate of 6000 f/m.

Removing the external trigger signal connector will automatically switch the *TRACKER* back to Internal Mode.

LOW BATTERY:

When the *TRACKER* detects that the battery charge is low, it flashes the LED display as a warning. This gives the user a chance to change batteries.



When the external trigger rate exceeds the maximum flash rate of the TRACKER [6000 f/m], missed flashes will result.

NOTE

IV. INTERFACE

The **TRACKER** utilizes two separate display panel connectors for power input and trigger input connections [Ref [Display Panel](#) - pg. 10]

Power Input Connector Pin Assignments:

2.1mm / 5.5mm Power Jack (-) ———— (●) ———— (+)

•Supply Input: +12VDC @ 1.5A, center pin (+)

NOTE: The **TRACKER** chassis is electrically connected to supply return (-).

Trigger Input Connector Pin Assignments:

Pin 1 •Pulse input: +2.5V pk min to +15V pk max, zero based pulse, 100msec min to 1msec max pulse width, 470 ohm input impedance.

Pin 2 •Return for pulse input signal.

NOTE: Pins 1 & 2 are electrically isolated from all **TRACKER** circuits, including chassis ground. [Optoisolator input]

For [Contact Closure](#) or [Open Collector](#) [sink] type signals, connect as follows:

•Pin 1 to 4. [Trigger supply]

•Pin 2 to Sink Signal. [[Contact Closure](#) or [Open Collector](#), 12VDC @ 30 mA min rating]

•Pin 5 to Sink return.

Pin 3 •External Trigger Select: Selects the external trigger mode when connected to trigger supply Pin 4.

Pin 4 •Trigger supply: 12VDC nominal, trigger supply is the same voltage supplied to the power input connector.

Pin 5 •Signal ground: Return for trigger supply.



NOTE

*The **TRACKER** external trigger cables contain an internal jumper from Pin 3 to Pin 4. This internal connection automatically selects external mode of operation.*

V. SPECIFICATIONS

Power Input: ·12 to 14.5 Volts DC; 13 watts max

Trigger:

External

Optoisolated: ·Pulse [Source] Input: +2.5V min to 15V max, zero based
100µsec to 1msec pulse width, 470 ohm input impedance.

Non-Isolated: ·Contact closure or open collector [sink] signal: 12VDC
@ 30mA; 100µsec to 1msec width.

Internal:

·Microcontroller timed 30-6000 (*standard*)

Trigger Supply: ·12 to 16 Volts DC internally connected to power input.

Energy Output: ·0.6477 joules/flash @ 3000 f/m
·0.2115 joules/flash @ 6000 f/m

Energy output automatically decreases from its maximum value to the value specified at the maximum flash rate.

Light Output:

Illumination: ·6950 Lux at 0.3 meters 100% intensity at 3000 f/m
·2980 Lux at 0.3 meters 100% intensity at 6000 f/m

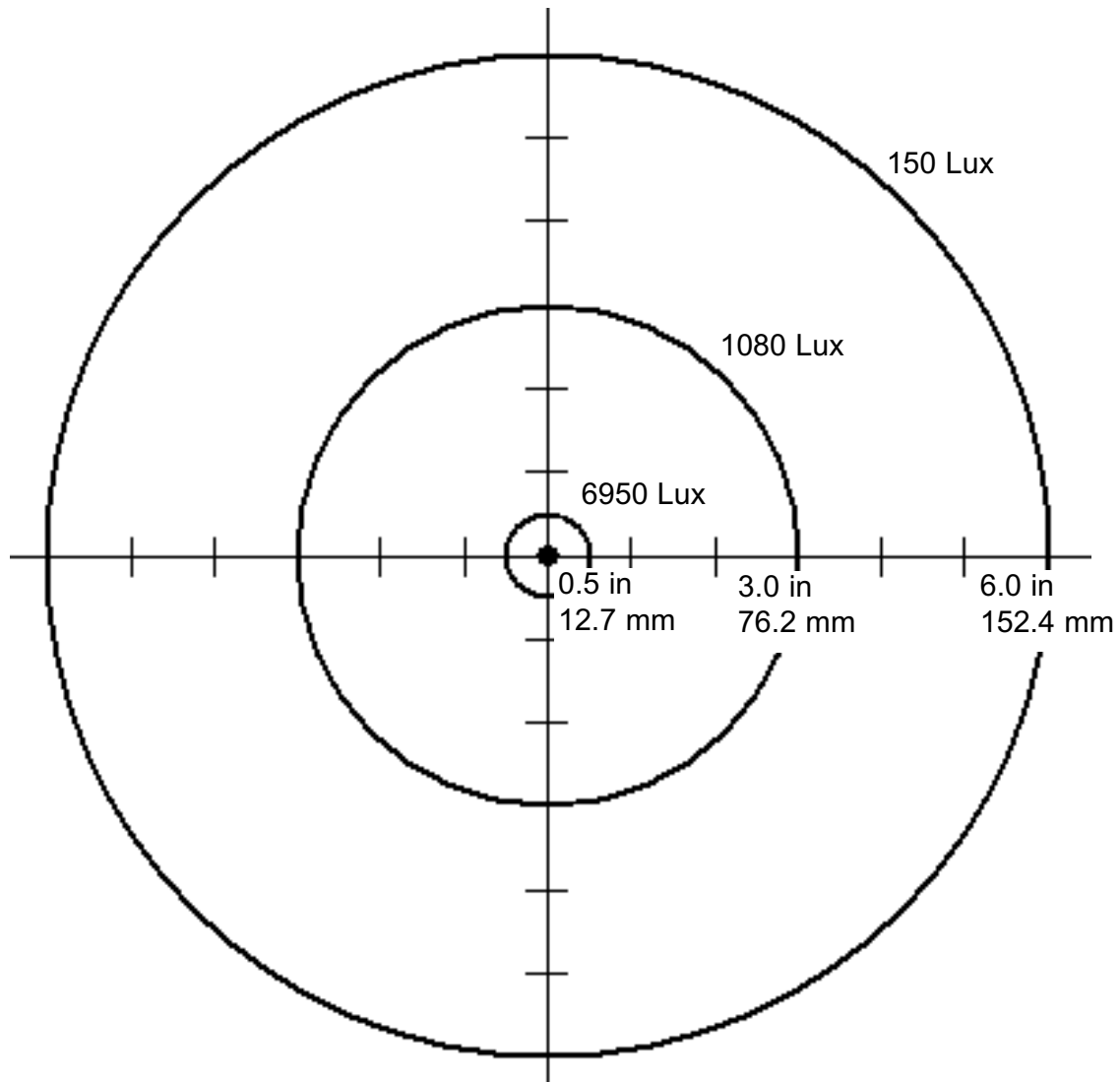
Flash Duration: ·25 microseconds

Physical: ·Dimensions: 7.62” [lg] x 4.31” [w] x 11.87” [ht], 4.16 lb.
184.1mm x 109.5 x 301.6mm, 1.88kg.
Dimensions were taken with the new DC Power handle and DC power pack.

Environmental: ·Continuous operating temperature range:
0°C to 40°C [32°F to 104°F]
·Humidity: 95% noncondensing

PHOTOMETRY

The isolux chart below shows the **TRACKER** illumination pattern at a target distance of 1 [one] foot.



NOTES:

- The distance increments shown above represent displacement from a target center
- Flash rate set to 3000 f/m [flashes per minute].

VI. CHANGING LAMP

The **TRACKER** lamp will need to be replaced whenever the lamp reaches its total end of life point. [ie. complete loss of flashing]

Before reaching its total end of life point, the lamp may exhibit reduced light output, as well as inconsistent flashing. Lamp replacement for this condition is recommended if the actual inspection application requires peak lamp performance.



WARNING! PROCEED WITH CAUTION

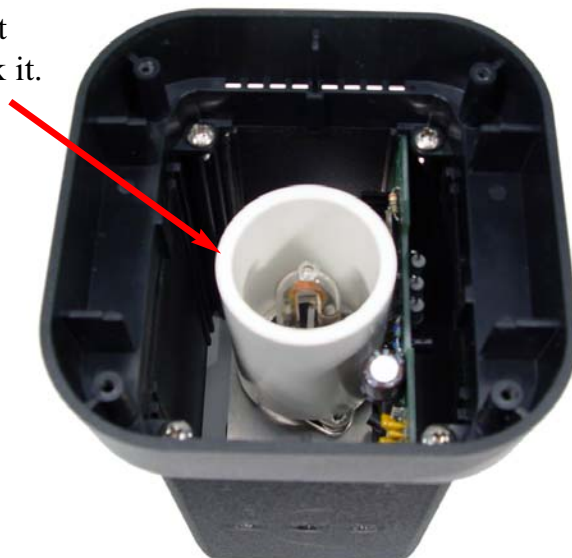
The TRACKER lamp operates at **LETHAL high voltages**. The lamp may be extremely **HOT** after only a few minutes of operation. Before opening, disconnect power supply and wait one minute to allow voltage to bleed off. Allow to **cool** as necessary.

1. Disconnect power supply cord.
2. Remove the window by unscrewing the four [4] truss head phillips screws.
3. Remove the reflector to expose the entire lamp.
4. Using the “**Lamp Removal Tool**”, supplied with your **TRACKER**, place it over the lamp, keeping the cutouts of the tool in line with the ears of the lamp flange.
5. Once the tool is seated, rotate it clockwise until it stops, then carefully pull the lamp from its socket and discard.
6. Unpack the new lamp. Using a clean dry cloth to prevent touching the lamp glass, place the new lamp into the removal tool in a locked position, and then line up the ‘key’ on the lamp center pin with the keyway in the lamp socket inside the **TRACKER**, and carefully push the lamp until it stops.
7. Turn the removal tool counter-clockwise to disengage it from the lamp.
8. Reverse steps 3 to 1 to re-assemble the unit.

**LAMP REMOVAL TOOL
62-1213**



Once the tool is seated flat, turn it clockwise to lock it.



VIII. CHARGING THE DC POWER PACK

To Charge the DC Power Pack

1. Press the release buttons on the DC Power Pack and firmly pull the Pack out of the DC Power handle.



2. Connect charger to an appropriate outlet before inserting DC Power Pack.



3. Insert the DC Power Pack into the charger as shown above. The **red** (charging) light will blink continuously indicating that the charging process has started.

- Charge time is approximately 1 hour.

4. The completion of charge will be indicated by the **red** light remaining ON continuously.
5. The pack is fully charged and may be used at this time or left in the charger.
CAUTION: MAINS voltage present at charging terminals. Do not probe with conductive objects. ***Danger of electric shock or electrocution may result.***



Note

To minimize the possibility of reduced DC Pack capacity due to the NiCd cell memory effect, do not charge the DC Power Pack until the low battery indicator is illuminated.

CHARGER INDICATORS

PACK CHARGING.....- - - - -
FULLY CHARGED.....
HOT/COLD PACK DELAY.....- . - . - . - . - . - . - . - . - .
REPLACE PACK......
PROBLEM POWER LINE......

HOT/COLD PACK DELAY

When the charger detects a DC Power Pack that is hot or cold, it automatically starts a Hot/Cold Pack Delay, suspending charging until the pack temperature has stabilized. After the DC Power Pack cooled, the charger automatically switches to the Pack Charging mode. This feature ensures maximum battery life. The red light flashes long, then short while in Hot/Cold Delay mode.

REPLACE PACK

The charger is designed to detect certain problems that can arise with DC Power Packs which would be indicated by the red light flashing at a fast rate. If this occurs, reinsert DC Power Pack. If problem persists, try a different pack to determine if charger is OK. If the new pack charges correctly, then the original pack is defective and should be returned to Unilux for recycling. If the new DC Power Pack elicits the same trouble indication as the original, have charger tested by an authorized Unilux service center.

PROBLEM POWER LINE

If the input voltage is out of tolerance, the charger may temporarily suspend operation and begin flashing the red light (two fast blinks followed by a pause). This indicates the power source is out of limits. If the voltage returns to an acceptable level, the charger will automatically switch to a normal charging mode and flash accordingly.

LEAVING THE DC POWER PACK IN THE CHARGER

The charger and DC Power Pack can be left connected with the red light glowing indefinitely. The charger will keep the DC Power Pack fresh and fully charged.

IMPORTANT CHARGING NOTES

1. Longest life and best performance can be obtained if the DC Power Pack is charged when the air temperature is between 65°F and 75°F (18°C - 24°C). **DO NOT** charge the DC Power Pack in an air temperature below +40°F (+4.5°C), or above +105°F (+40.5°C). This is important and will prevent serious damage to the DC Power Pack.
2. The charger and DC Power Pack may become warm to touch while charging. This is a normal condition, and does not indicate a problem.
3. If the DC Power Pack does not charge properly, (1) Check power at receptacle by plugging in another piece of equipment, (2) Move charger and pack to a location where the surrounding air temperature is approximately 65°F - 75°F (18°C - 24°C), (3) If charging problems persist, take or send the equipment, DC Power Pack and charger to an authorized Unilux service center.
4. The DC Power Pack should be recharged when it fails to produce sufficient power enough to keep the equipment functioning properly. **DO NOT CONTINUE** to use under these conditions. Follow the charging procedure. You may also charge a partially used pack whenever you desire with no adverse affect on the DC Power Pack.
5. Under certain conditions, with the charger plugged in to the power supply, the exposed charging contacts inside the charger can be shorted by foreign material. Foreign materials of a conductive nature such as, steel wool, aluminum foil, or any buildup of metallic particles should be kept away from charger cavities. Always unplug the charger from the power supply when there is no DC Power Pack in the cavity. Unplug the charger before attempting to clean.
6. Do not immerse charger in water or any other liquid.
7. For optimum performance, leave the DC Power Pack in the charger for six hours or more once every 10 to 20 recharges. This balances the cells, keeps the pack at peak performance and extends the usable life of the pack.

WARNING: Do not allow any liquid to get inside charger. Electric shock may result. To facilitate the cooling of the DC Power Pack after use, avoid placing the charger or pack in a warm environment.

CAUTION: Never attempt to open the DC Power Pack for any reason. If the plastic housing of the pack breaks or cracks, return the DC Power Pack to Unilux.

IMPORTANT SAFETY INSTRUCTIONS

FOR 14.4V 1 HOUR CHARGER

- This manual contains important safety and operating instructions for the Unilux DC Power Pack 1 Hour Charger.
- Before using the charger, read all instructions and cautionary markings on 1) charger, (2) DC Power Pack, and (3) product using DC Power Pack.
- **DANGER: LINE VOLTAGE PRESENT AT CHARGING TERMINALS.** Do not probe with conductive objects. Danger of electric shock or electrocution.
- **DANGER:** If DC Power Pack case is cracked or damaged, do not insert into charger. Danger of electric shock or electrocution.
- **DANGER:** Never attempt to open the DC Power Pack for any reason. If the plastic housing of the DC Power Pack breaks or cracks, return to Unilux for recycling.
- **DANGER:** Don't allow any liquid to get inside charger. Electric shock may result. To facilitate the cooling of the DC Power Pack after use, avoid placing the charger or DC Power Pack in a warm environment such as in a metal shed, or an uninsulated trailer.
- **CAUTION:** To reduce risk of injury, charge only Unilux rechargeable DC Power Packs. Other types of Power Packs may burst, causing personal injury and damage.
- Do not expose charger to rain or snow.
- Use of an attachment not recommended or sold by Unilux may result in a risk of fire, electric shock, or injury to persons.
- These chargers are not intended for any uses other than charging Unilux rechargeable DC Power Packs. Any other uses may result in a risk of fire, electric shock, or electrocution.
- To reduce risk of damage to electric plug and cord, pull by plug rather than cord when disconnecting charger.
- Make sure cord is located so that it will not be stepped on, tripped over, or otherwise subjected to damage or stress.
- An extension cord should not be used unless absolutely necessary. Use of improper extension cord could result in risk of fire, electric shock, or electrocution.

IMPORTANT SAFETY INSTRUCTIONS



WARNING: When using electrically operated equipment, basic safety precautions should always be followed to reduce risk of fire and personal injury.

READ ALL INSTRUCTIONS

- Do not incinerate the DC Power Pack even if it is severely damaged or is completely worn out. The DC Power Pack can explode in a fire.
- A small leakage of liquid from the DC Power Pack cells may occur under extreme usage or temperature conditions. This does not indicate a failure. However, if this leakage gets on your skin:
 - a. Wash quickly with soap and water.
 - b. Neutralize with a mild acid such as lemon juice or vinegar.
 - c. If the DC Power Pack liquid gets into your eyes, flush them with clean water for a minimum of 10 minutes and seek immediate medical attention. (MEDICAL NOTE: The liquid is 25%-35% solution of potassium hydroxide.)
- Never attempt to open the DC Power Pack for any reason. If the plastic housing of the DC Power Pack breaks or cracks, immediately discontinue use and do not recharge.

DC PACK CELLS

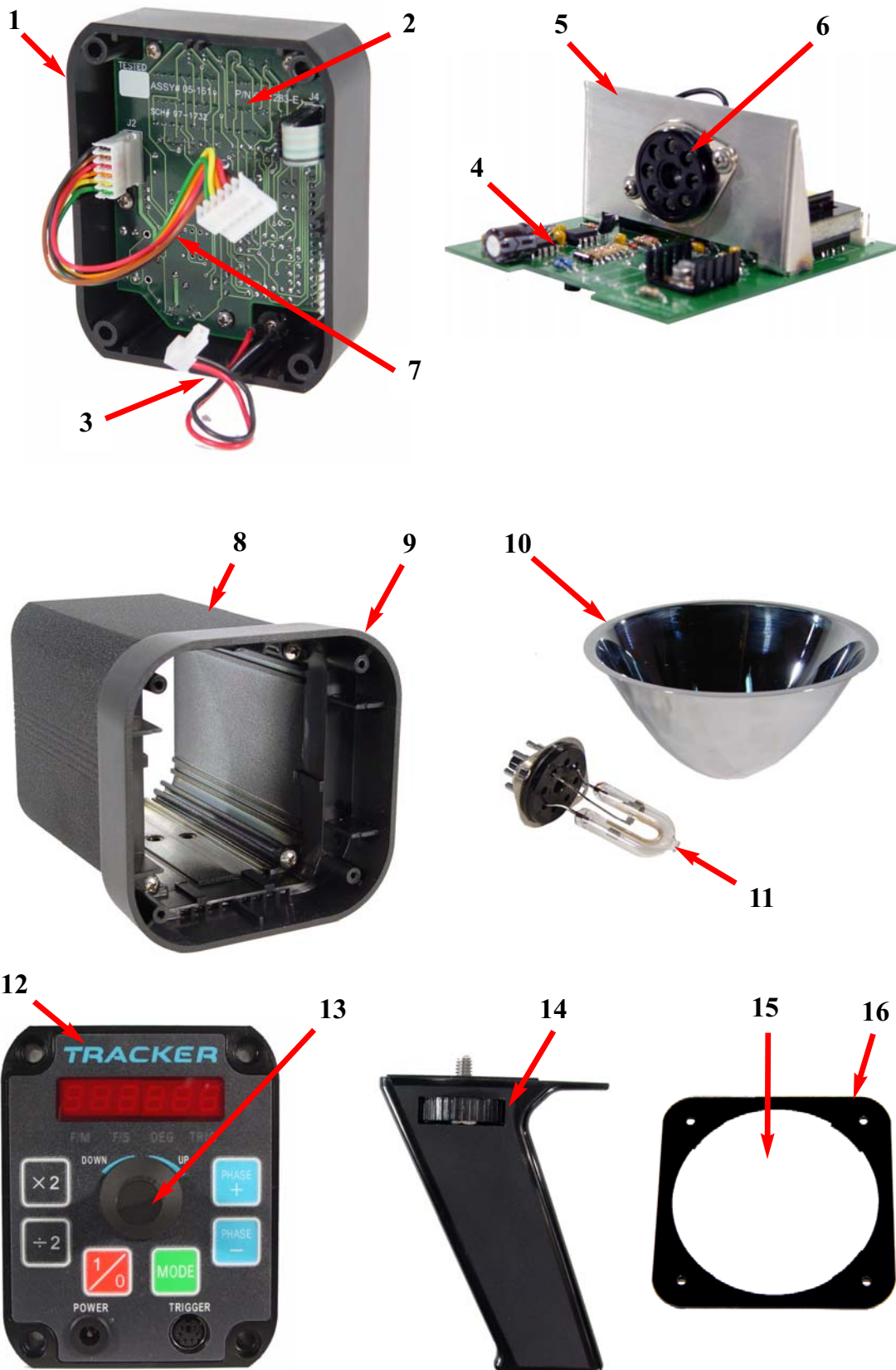
After several years of normal use, the cells in your DC Power Pack will no longer accept a charge. This is a normal occurrence, and signifies that the DC Power Pack cells have reached the end of their useful life. Replacement DC Power Packs are available from Unilux, order part No. UPN 77-1055.

Your DC Power Pack contains nickel cadmium cells. The Environmental Protection Agency considers cadmium to be a toxic material that can do severe damage to the environment. Check with your state Environmental Protection Agency to find out how to properly recycle or dispose of your DC Power Pack.

VIII. SPARE PARTS & ACCESSORIES

DESCRIPTION	UNILUX PART NUMBER
Flashlamp-----	45-1046
Window-----	04-1569
<u><i>Wall Plug-In Power Supply</i></u>	
USA - 120 VAC to 12 VDC - 25 Watt-----	56-1034
European - 240 VAC to 12 VDC - 25 Watt-----	56-1035
UK - 240 VAC to 12 VDC - 25 Watt-----	56-1036
Australia - 240 VAC to 12 VDC - 25 Watt-----	56-1037
<u><i>Fixed Mounting</i></u>	
45" Articulating Arm-----	04-1633
<u><i>Mounting Brackets [for Articulating Arm]</i></u>	
Clamp Mounting Bracket-----	56-1006
Wall Mounting Bracket-----	56-1007
Surface Mounting Bracket-----	56-1008
Pulse Input Cable - 15 ft. [4.5M]-----	04-1427
Contact Closure Cable - 15 ft. [4.5M]-----	04-1428
Thru Scan Optical Pickup Cable 15 ft. [4.5M]-----	04-1343-4
Battery Pack Handle Kit-----	04-1597-110
<u><i>consists of:</i></u>	
DC Power Pack -----	77-1055
1 Hour Charger 110V-----	56-1040-110 (*See Note)
DC Power Handle -----	04-1595
Mounting plate -----	81-1985
(2) #10 100Deg. Flat HD Phil. Screw-----	90-1210
 *Note: Different 1 Hour Chargers are available:	
1 Hour Charger 220V Europe-----	56-1040-220
1 Hour Charger 220V Australia-----	56-1040-220AU
1 Hour Charger 220V UK -----	56-1040-220UK

TRACKER ITEMIZED



TRACKER ITEMIZED PARTS LIST

REF NO.	PART NO.	DESCRIPTION
1	63-1055	REAR COVER
2	05-1611	CONTROL/DISPLAY PCB
3	05-1622	DC POWER JACK HARNESS
4	05-1573	POWER PCB
5	81-1968	LAMP SOCKET SUPPORT PLATE
6	40-1631	LAMP SOCKET
7	04-1642	CONTROL CABLE
8	85-1248	CHASSIS
9	63-1054	REFLECTOR/WINDOW BEZEL
10	88-1209	REFLECTOR
11	45-1046	FLASH LAMP
12	70-1184	KEYPAD
13	96-1267	RATE CONTROL KNOB
14	05-1559	HANDLE
15	88-1201	WINDOW
16	63-1056	WINDOW BACKER

IX. SERVICE and WARRANTY

LIMITED WARRANTY

All parts of the **TRACKER**, with the exception of the lamp and trigger transformer are warranted against defects of materials and workmanship for a period of one [1] year from date of delivery.

Unilux makes no other warranties, expressed, implied, or of merchantability for this equipment. If the **TRACKER** or accessory does not function properly during the warranty period, Unilux will repair it without charge according to the terms stated. Unilux will not be responsible for any consequential or incidental damages resulting from the sale, use or improper functioning of this equipment even if loss or damage is caused by the negligence or fault of Unilux.

SERVICE PROCEDURE UNDER WARRANTY

The **TRACKER** lamp and electronics operate at **LETHAL high voltages**. Refer all repairs to Unilux service department.

If the unit or component must be returned to Unilux for repair or replacement, contact Customer Service for a Return Authorization Number [RA#].

After receipt of the Return Authorization Number, pack the unit properly and ship it to Unilux, (prepaid) with the RA# clearly marked on the outside of the package. Unilux will repair or replace the item and return it prepaid.

Servicing

- As stated in the operation section re: **High Voltage**, servicing this equipment beyond changing the flashlamp is not recommended. For repair, consult the factory in the U.S.A.

UNILUX INC.

59 NORTH 5TH STREET
SADDLE BROOK N.J. 07663 U.S.A.



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